AQ1.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.
3.1 Level 1:

Consider a system of linear equations (System 1):

$$\begin{aligned} -2x_1 + 3x_2 + x_3 &= 1 \\ -x_1 + x_3 &= 0 \\ 2x_2 &= 5 \end{aligned}$$

Answer questions 1 and 2 based on the above data.
1 point

If the matrix representation of system (1) is $Ax = b$, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, then

- [x] $A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$
- [ ] $b = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$
- [ ] $A = \begin{bmatrix} -2 & -1 & 0 \\ 3 & 0 & 2 \\ 1 & 1 & 0 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$
- [x] $A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$

Yes, the answer is correct.
Score: 1
Targeted Feedback:

-

Feedback:

Accepted Answers:

$$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}, \text{ and } b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$$

1 point

System (1) has

[Hint: Solve for $x_1, x_2,$ and x_3.]

- (●) a unique solution.
- ( ) no solution.
- ( ) infinitely many solutions.
- ( ) None of the above.

Yes, the answer is correct.

Score: 1

Accepted Answers:

a unique solution.

1 point

Choose the set of correct options.

[Hint: Think of A as a 2×2 or 3×3 matrix, and b accordingly.]

- [x] Every system of linear equations has either a unique solution, no solution or infinitely many solutions.
- [ ] If each equation of a system of linear equations is multiplied by a non-zero constant c, then the solution of the new system of equations is c times the solution of the old system of equations.
- [x] If $Ax = b$ is a system of linear equations which has a solution, then the system of linear equations $cAx = b$, where $c \neq 0$, will also have a solution.
- [x] If $Ax = b$ is a system of linear equations which has a solution, then $\frac{1}{c}Ax = b$, where $c \neq 0$, will also have a solution.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Every system of linear equations has either a unique solution, no solution or infinitely many solutions.

If $Ax = b$ is a system of linear equations which has a solution, then the system of linear equations $cAx = b$, where $c \neq 0$, will also have a solution.

If $Ax = b$ is a system of linear equations which has a solution, then $\frac{1}{c}Ax = b$, where $c \neq 0$, will also have a solution.

1 point

The Plane 1 and Plane 2 in Figure M2W1AQ3, correspond to two different linear equations, which form a system of linear equations.

The above system of linear equations has

[Hint: If a system of linear equations has a solution, then the point corresponding to the solution must lie on each plane corresponding to each linear equation of the given system.]

- ( ) a unique solution.
- (●) no solution.
- ( ) infinitely many solutions.
- ( ) None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
no solution.

1 point

Consider the geometric representations (Figures (a), (b), (a), (b), and $(c$(c)) of three systems of linear equations.

Choose the set of correct options.

- [x] Figure (a)(a) represents a system of linear equations which has no solution.
- [ ] Figure (a)(a) represents a system of linear equations which has infinitely many solutions.
- [x] Figure (b)(b) represents a system of linear equations which has a unique solution.
- [ ] Figure (b)(b) represents a system of linear equations which has infinitely many solutions.
- [x] Figure (c)(c) represents a system of linear equations which has infinitely many solutions.
- [ ] Figure (c)(c) represents a system of linear equations which has no solution.

Yes, the answer is correct.
Score: 1
Accepted Answers:
Figure (a)(a) represents a system of linear equations which has no solution.
Figure (b)(b) represents a system of linear equations which has a unique solution.
Figure (c)(c) represents a system of linear equations which has infinitely many solutions.

3.2 Level 2:

1 point

Consider a system of equations: $\begin{aligned} 2x_1 + 3x_2 &= 6 \\ -2x_1 + kx_2 &= d \\ 4x_1 + 6x_2 &= 12 \end{aligned}$ 2x1 + 3x2−2x1 + kx24x1 + 6x2 = 6 = d = 12Choose the set of correct options.

- [ ] $Ax = b$Ax = b represents the above system, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$x = [x1x2], $A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$A = [2−243k6], and $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$b = [6d12]
- [ ] The system has no solution if $k = -3$k = −3, $d = 0$d = 0.
- [ ] The system has a unique solution if $k = 3$k = 3, $d = 0$d = 0.

☐
The system has infinitely many solutions if $k = -3$, $d = 6$.
☐
The system has infinitely many solutions if $k = -3$, $d = -6$.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$Ax = b$ represents the above system, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$,

and $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$

The system has no solution if $k = -3$, $d = 0$.
The system has a unique solution if $k = 3$, $d = 0$.
The system has infinitely many solutions if $k = -3$, $d = -6$.

1 point
Let x_1 and x_2 be solutions of the system of linear equations $Ax = b$. Which of the following options are correct?

☐
$x_1 + x_2$ is a solution of the system of linear equations $Ax = b$.
☐
$x_1 + x_2$ is a solution of the system of linear equations $Ax = 2b$.
☐
$x_1 - x_2$ is a solution of the system of linear equations $Ax = b$.
☐
$x_1 - x_2$ is a solution of the system of linear equations $Ax = 0$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$x_1 + x_2$ is a solution of the system of linear equations $Ax = 2b$.
$x_1 - x_2$ is a solution of the system of linear equations $Ax = 0$.

1 point
Let v be a solution of the systems of linear equations $A_1 x = b$ and $A_2 x = b$. Which of the following options are correct ?

☐
v is a solution of the system of linear equations $(A_1 + A_2)x = b$.
☐
v is a solution of the system of linear equations $(A_1 + A_2)x = 2b$.
☐
v is a solution of the system of linear equations $(A_1 - A_2)x = 0$.
☐
v is a solution of the system of linear equations $(A_1 - A_2)x = b$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
v is a solution of the system of linear equations $(A_1 + A_2)x = 2b$.
v is a solution of the system of linear equations $(A_1 - A_2)x = 0$.

Consider a system of equations: $\begin{aligned} x_1 - 3x_2 &= 4 \\ 3x_1 + kx_2 &= -12 \end{aligned}$x1 − 3x23x1 + kx2 = 4 = −12Where $k \in R$k ∈ R.

If the given system has a unique solution, then kk should not be equal to

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -9
1 point
Consider two system of linear equations:
System 1:

$$\begin{aligned} x_1 + 2x_2 &= -5 \\ 0x_1 - x_2 &= 5 \end{aligned}$$x1 + 2x20x1 − x2 = −5 = 5

System 2:

$$\begin{aligned} -2x_3 + x_4 &= -5 \\ 3x_3 + x_4 &= 5 \end{aligned}$$−2x3 + x43x3 + x4 = −5 = 5

Suppose there is another system of linear equation given by
$$\begin{aligned} (1-2)(x_1 + x_3) + (2+1)(x_2 + x_4) &= m \\ (0+3)(x_1 + x_3) + (-1+1)(x_2 + x_4) &= n \end{aligned}$$(1 − 2)(x1 + x3) + (2 + 1)(x2 + x4)(0 + 3)(x1 + x3) +

(−1 + 1)(x2 + x4) = m = n
for some real values of mm and nn. Find the value of $n - m$n − m.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 46
1 point